

Alejandro Roa Aparicio

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PROFILE

- » Mechatronics Engineer and Software Engineer
- » Focus on robotics and applied artificial intelligence, from physical integration and computer vision to automation and scalable cloud solutions

EDUCATION

- 📅 2027 **Mechatronics Engineering**
Universidad EAN
📍 Bogotá, Colombia
» Degree Project: Armando - Firearm detection system using computer vision with alerts to authorities in Colombia
- 📅 2026 **Software Engineering**
Universidad EAN
📍 Bogotá, Colombia
» Degree Project: Robot Angel - Open Source Robotics IDE
- 📅 2021 **Technical High School Specialized in Mechatronics**
Instituto Técnico Industrial Francisco José De Caldas
📍 Bogotá, Colombia
» Technical training in Mechatronics

WORK EXPERIENCE

- 📅 06/2026 - present **Technology Intern**
DAEM - ESP OUTSOURCING
📍 Bogotá, Colombia
» Supporting the technology area in the development of a multi-agent legal RAG
- 📅 01/2026 - 06/2026 **Research Intern**
Innovation and Transfer Management, Universidad EAN
📍 Bogotá, Colombia
» Developed an environmental monitoring system with real-time visualization and prediction for Universidad EAN's research management
- 📅 01/2026 - 03/2026 **Junior Programmer**
WhaleDog
📍 Bogotá, Colombia
» Full stack development for a senate candidate's campaign. Responsible for deployment, design, DevOps, frontend and backend
- 📅 09/2025 - 12/2025 **Junior Programmer**
Freelancer
📍 Bogotá, Colombia
» Developed a Linux simulation with Geolandspill for crude oil spill modeling for an oil company

📅 06/2025 - 09/2025 **Assistant Teacher**

Fundación Nuevo Amanecer ESAL Chocó and Universidad EAN

📍 Chocó, Colombia

» Taught classes in the STEMChocó project, through which we built a CANSAT with low-income girls

📅 06/2025 - 12/2025 **University Tutor - Circuits and Electronics**

Universidad EAN

📍 Bogotá, Colombia

» Provided tutoring and collaborated in directing class laboratories for 2 subjects: Electronics and Circuits

TECHNICAL SKILLS

- | | | |
|------------|----------------------|-------------------|
| ✓ Robotics | ✓ Web Development | ✓ 3D Design |
| ✓ Python | ✓ Mobile Development | ✓ Circuit Design |
| ✓ C/C++ | ✓ Computer Vision | ✓ Prototyping |
| ✓ ROS | ✓ AI/ML/DL | ✓ Cloud Computing |
| ✓ Git | ✓ Linux | ✓ Docker |

FEATURED PROJECTS

- » **Corto Circuito - Science Outreach** Science outreach project where I share tips, explain concepts and projects, and seek to convey my love for science and engineering. [📷 @corto.circuitx](#)
- » **Robot Angel - Open Source Robotics IDE** Systems engineering degree project with the goal of becoming the #1 robotics-dedicated IDE worldwide. 100% open source project to democratize robotics one dreamer at a time. [🌐 Website](#) [🐙 GitHub](#)

OTHER PROJECTS AND PORTFOLIO

- » To see all my projects, competition robots, web development work, 3D designs, embedded systems and more, visit my complete portfolio:
- » 🌐 <https://alejandrora.engineer/>

LANGUAGES

Spanish (native)

English (B2)

COMPETITIONS AND HACKATHONS

- » Throughout my career, I have participated in various competitions and hackathons that have challenged my technical skills and teamwork abilities.
- | | |
|--|---|
| ✓ SostechPreneur 2025 Hackathon EAN | ✓ Engineering Faculty Hackathon EAN University 2023 |
| ✓ RHA International Robotics Battle 1L | ✓ Engineering Faculty Hackathon EAN University 2024 |
| ✓ CANSAT International Congress of Engineering and Automation 2024 | ✓ EAN BOT 2025 |

BADGES

- » Professional certifications verified on Credly
- » [🌐 https://www.credly.com/users/alejandro-roa-aparicio](https://www.credly.com/users/alejandro-roa-aparicio)